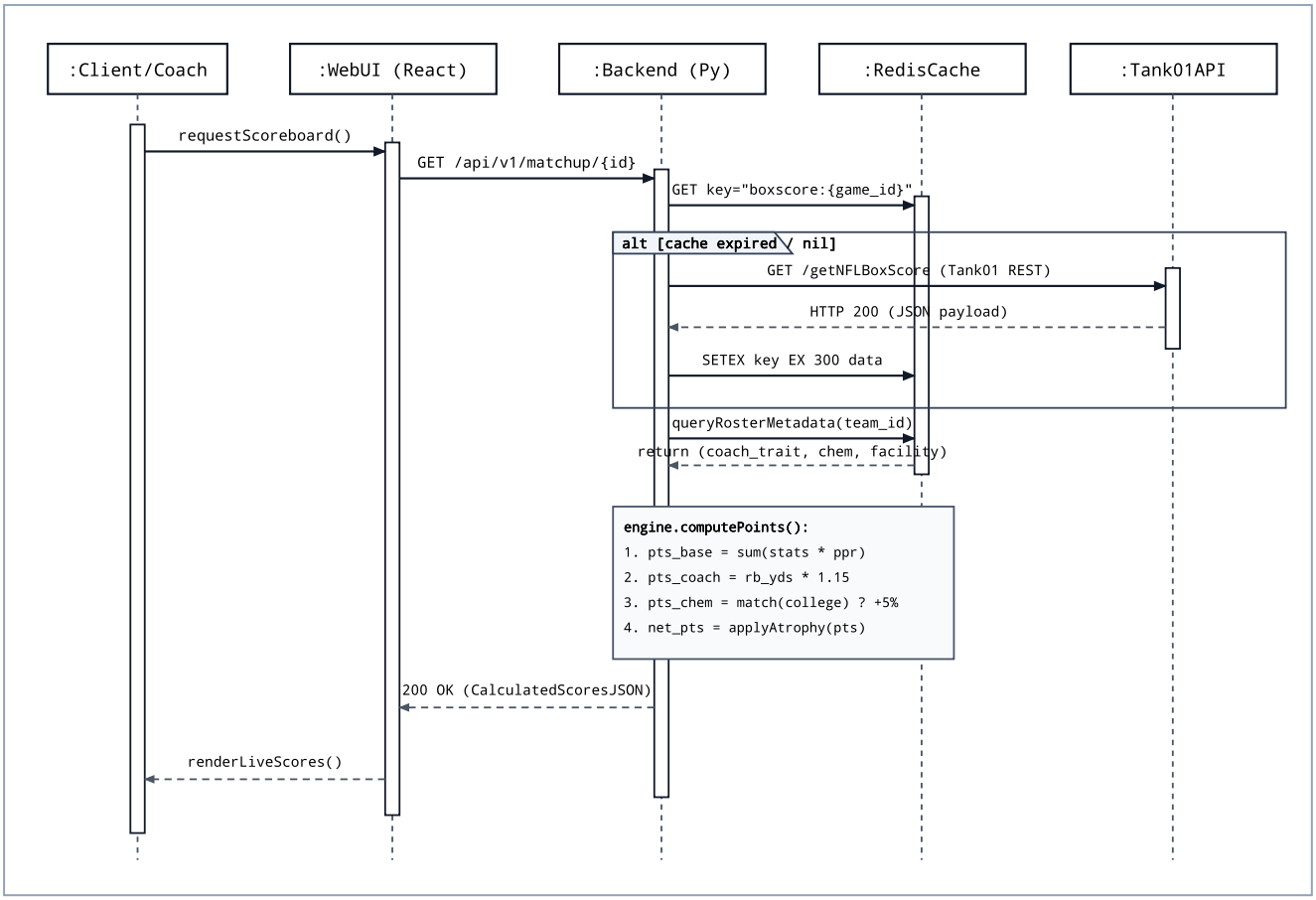
System Sequence Specification: Ultimate Fantasy Football (Group 10)

Authors: Miles Goldberger (MLG23C), Edward Payne (ECP23A), Tanner Simmonds (TKS23C), Amadou Landoure (ARL20BO)

1.0 Weekly Scoring Engine & Tank01 Box Score Ingestion

Figure 1.0: Live polling sequence during active game windows. Python backend intercepts client requests and queries Redis cache (TTL: 300s) to prevent exceeding Tank01 Basic Tier rate quotas (1,000 requests/month).



DESIGN CONSTRAINTS & ALGORITHMIC NOTES

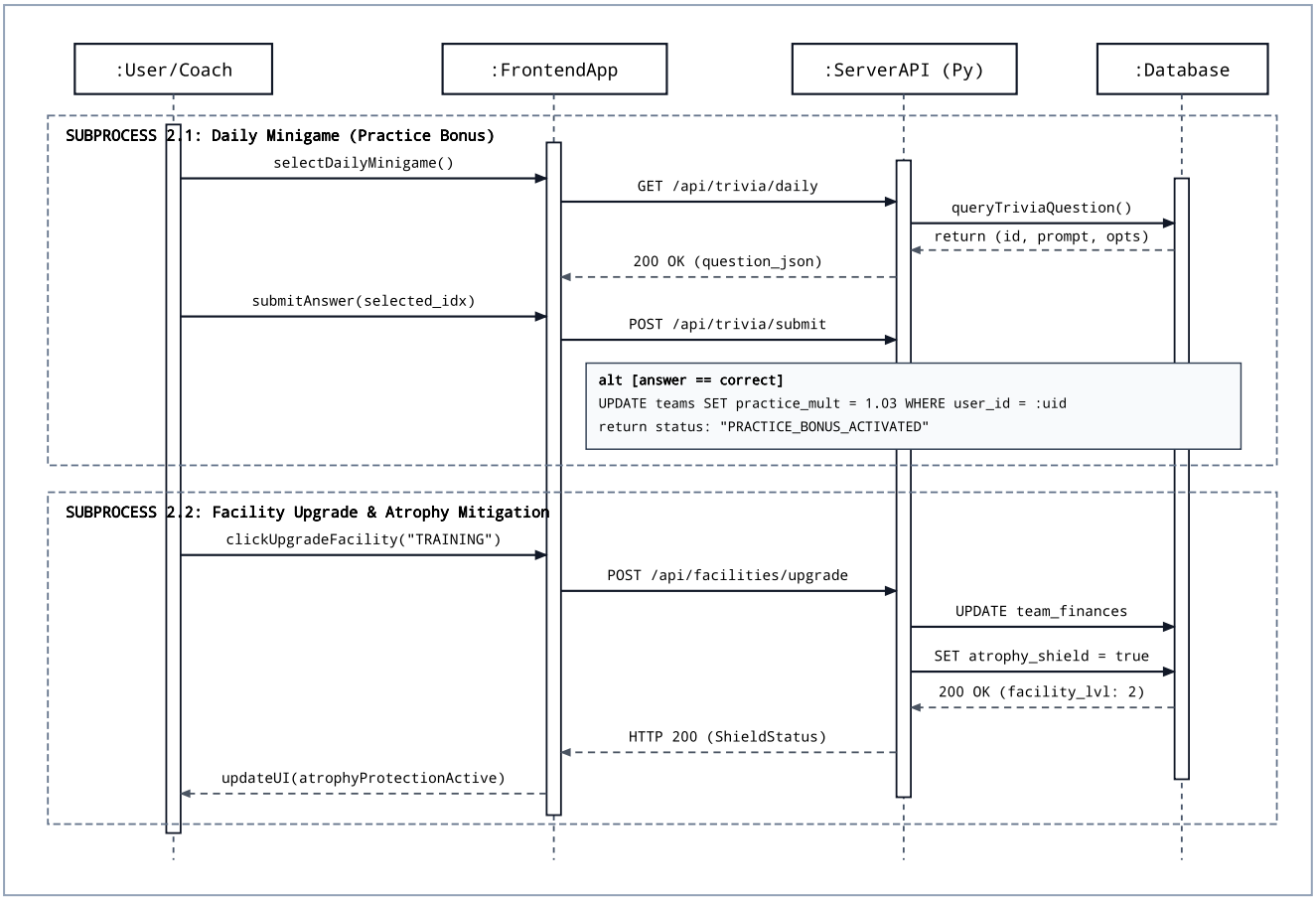
- **API Quota Budget:** RapidAPI Tank01 Basic provides 1,000 monthly calls. An aggressive 300-second (5 min) TTL prevents redundant fetches during concurrent user scoreboard traffic.
- **Execution Pipeline:** Multipliers are evaluated dynamically during aggregation; raw player statistical events remain canonical in PostgreSQL storage.

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2.0 Daily Trivia Minigame & Training Facility Subsystems

Figure 2.0: Subprocesses for weekday player engagement. Subprocess 2.1 verifies daily football trivia to award practice point multipliers; Subprocess 2.2 updates stadium and training facility state to prevent weekly stat degradation.



STATE LIFECYCLE SPECIFICATIONS

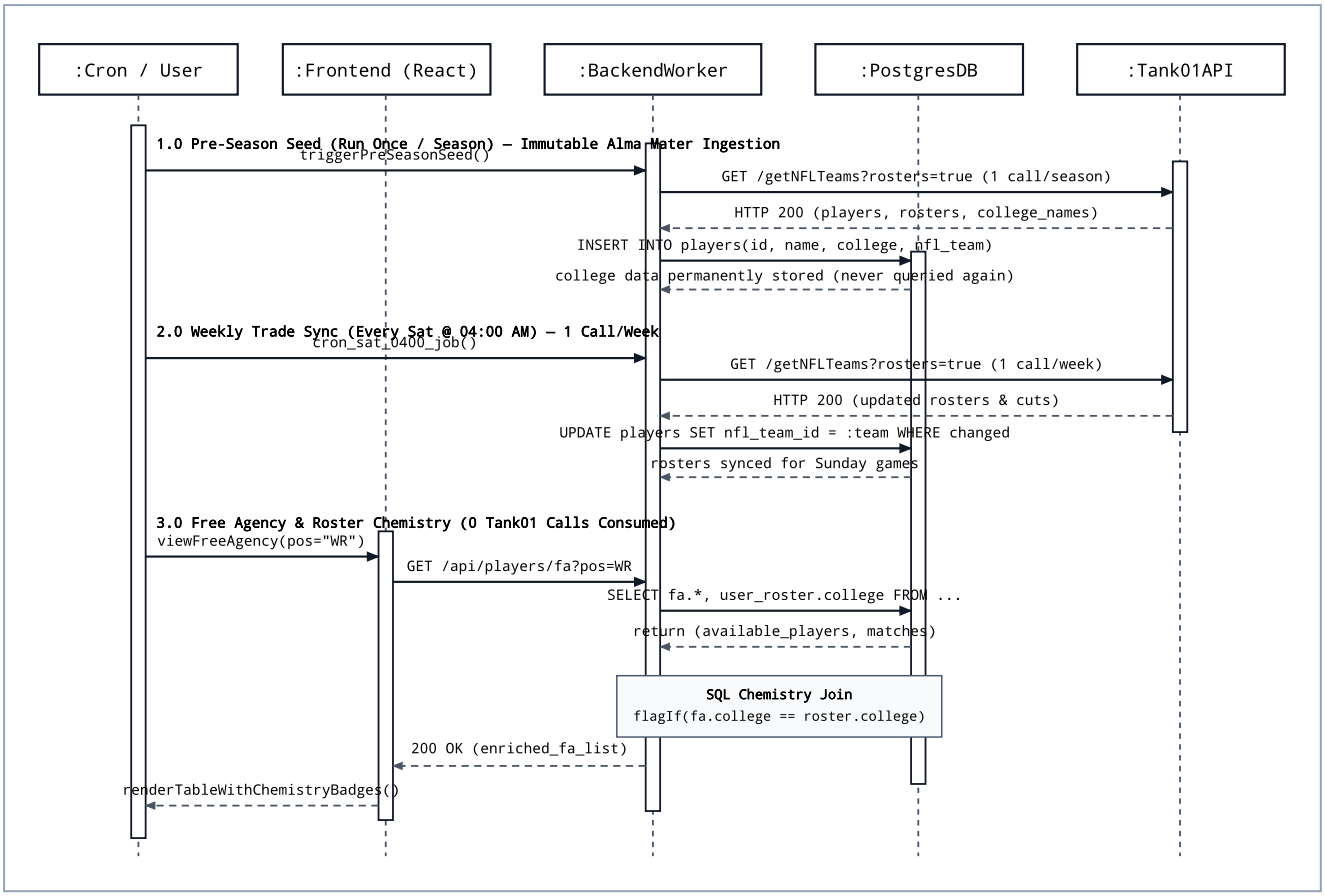
- **Practice Bonus Reset:** Practice multipliers granted from trivia expire weekly on Tuesday at 03:00 UTC (post-Monday Night Football stat finalization).
- **Atrophy Metric:** Active rosters experience a base 0.5% weekly performance decay if training facility maintenance is not maintained.

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3.0 Data Stratification Pipeline & Free Agency Chemistry Discovery

Figure 3.0: Tiered ingestion pipeline preserving the 1,000 monthly Tank01 API quota. Static college alma maters are seeded once per season; active NFL rosters are synchronized weekly on Saturday; user chemistry queries execute strictly against local PostgreSQL storage (0 external API calls).



API QUOTA ALLOCATION & STORAGE ARCHITECTURE

- **Immutable College Alma Mater:** College alma mater is fetched strictly once at pre-season initialization and stored permanently in `players.college`. No API calls are consumed for college data during the regular season.
- **Weekly Saturday Trade Sync:** Runs as a cron job every Saturday at 04:00 AM UTC via `getNFLTeams?rosters=true` (1 call/week = ~4 calls/month), updating trades, cuts, and depth charts prior to Sunday kickoff.
- **Monthly Tank01 Call Allocation:** Out of the 1,000 monthly limit, only 4–5 calls are allocated for roster synchronization, leaving **~995 calls exclusively for live game-day scoring** on Sundays/Mondays.